INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

SQUIB, MARK 1 MOD 0 FOR 2.75 INCH ROCKET

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense within the distribution limitations noted at the bottom of the page.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for one type of squib designated Mark 1 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this specification are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified required documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8410t), Indian Head, MD 20640-5035 by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-2550	Ammunition, General Specification for

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	Ammunition Lot Numbering
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVY (NAVAIR)

233AS522	Squib, Mark 1 Mod 0
233AS530	Casing Assembly
233AS531	Spacer Assembly
233AS532	Plug Assembly
233AS533	Initiation Assembly
233AS534	Bridge Wire Assembly
233AS814	Squib Closed Bomb Assembly

(Copies of the drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. Materials shall be in accordance with drawings and specifications specified herein.

3.3 Electrical continuity and resistance. The electrical circuit of each squib shall be continuous and shall meet the requirements of Drawing 233AS522.

3.4 Dielectric characteristics. The resistance between the connected lead wires and the case shall be not less than 50 megohms.

3.5 No-fire sensitivity. The squib shall not fire when energized with 0.20 amperes, 24 volts direct current (dc) for 5.25 ± 0.25 seconds.

3.6 Firing time. When subjected to 1.5 ± 0.15 amperes dc at 21.1 ± 2.8° Celsius (C), the squib shall fire in 0.005 seconds or less.

3.7 Pull test. The plug assembly (Drawing 233AS532) shall be of such quality as to withstand a 10 to 10.1 pound (lb) pull test on the individual leads.

3.8 Load test. The bridge wire assembly (Drawing 233AS534) shall have a sound electrical and mechanical joint that will not break when subjected to a 25 to 27 gram load test.

3.9 Output limits. A closed bomb test shall be used to measure the output pressure limits of the squib (see 4.9.7). When subjected to 1.5 ± 0.15 amperes dc firing current applied across the lead wires at 70 ± 5 degrees Fahrenheit (°F) (21.1 ± 2.8 degrees Celsius (°C)), the squib shall fire within 0.005 second or less of current application and the maximum output pressure produced by the squib shall be 220 through 455 pounds per square inch gage (psig).

3.10 Jolt resistance. The squib shall withstand the jolt test as specified in 4.9.8.1. Thereafter, it shall meet the requirements of 3.3 and 3.4 and shall fire in accordance with 4.9.4 in 0.005 seconds or less.

3.11 Vibration resistance. The squib shall withstand the vibration test as specified in 4.9.8.2. Thereafter, it shall meet the requirements of 3.3 and 3.4 and shall fire in accordance with 4.9.4 in 0.005 seconds or less.

3.12 Watertightness. The squib shall be watertight and shall show no evidence of leakage when tested as specified in 4.9.8.3 and thereafter shall fire in accordance with 4.9.4 in 0.005 seconds or less.

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3.13 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, metal shavings, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Known item defects shall not be submitted with the lot.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with 4.6.2.

4.3 First article inspection.

4.3.1 First article sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 1000 assemblies and 50 complete sets of parts (i.e., 50 each of every component, and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials, (see 4.5.c.). A sample containing known defects will not be submitted unless specifically authorized, by the contracting officer. A first article, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period greater than 90 days or whenever a change occurs in manufacturing process, materials used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings (see figure 1).

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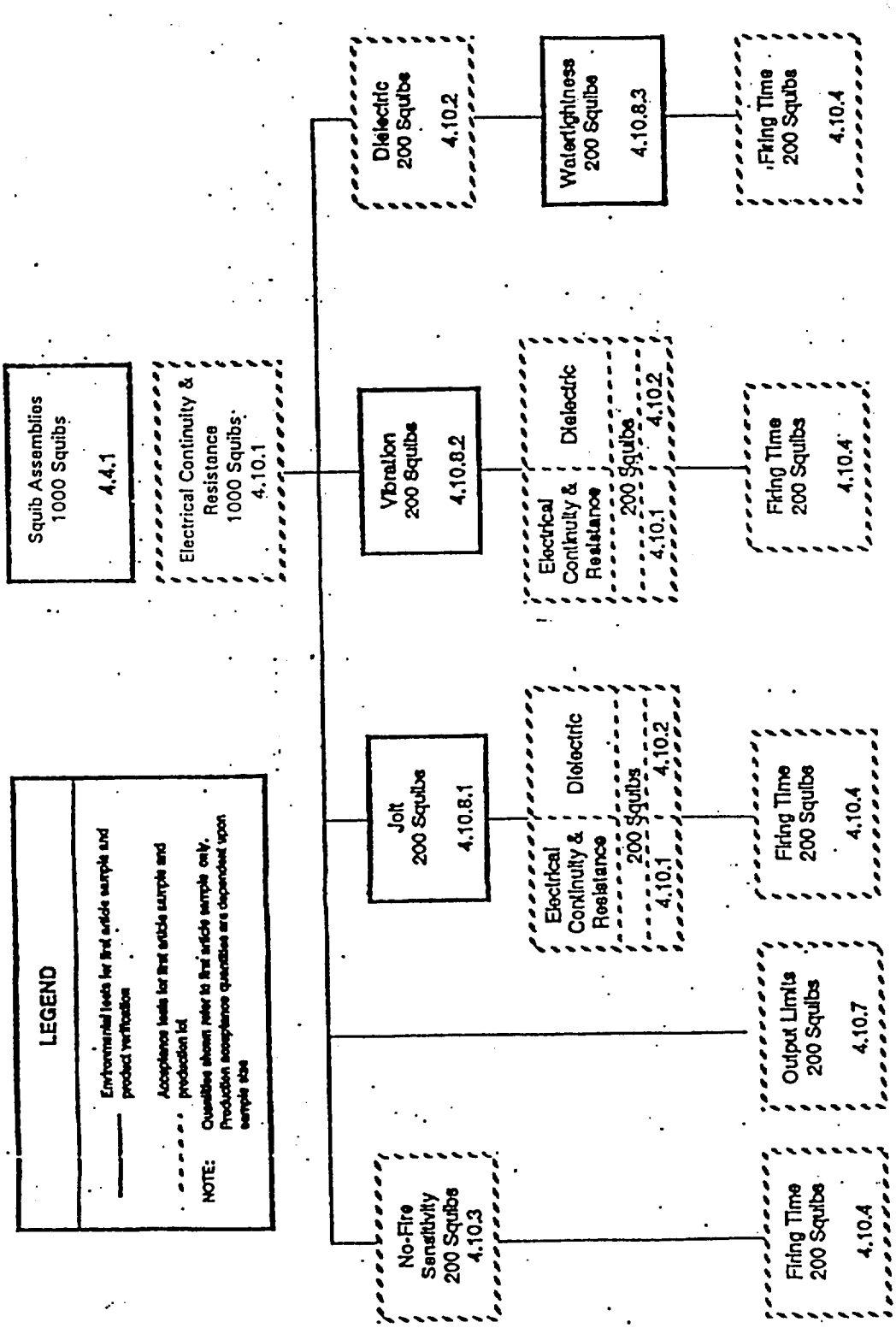


FIGURE 1. Test sequence.

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4.3.2.1 Sample. Fifty percent of the squibs for the first article testing shall demonstrate operability when tested at -53.9 ± 2.8 °C and the remaining 50 percent when tested at 73.9 ± 2.8 °C.

4.3.3 Rejection. If any assembly or component fails to comply with any of the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or component to comply with any of the stated requirements.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in table I, the applicable drawings as specified herein, and as specified by the contracting officer (see 6.2).

TABLE I. *Quality conformance inspection.*

Inspection	Requirement paragraph	Test paragraph
Material	3.2	4.5
Electrical continuity and resistance	3.3	4.8.1
Dielectric characteristics	3.4	4.8.2
No-fire sensitivity	3.5	4.8.3
Firing time	3.6	4.8.4
Pull test	3.7	4.8.5
Load test	3.8	4.8.6
Output limits	3.9	4.8.7
Jolt resistance	3.10	4.8.8.1
Vibration resistance	3.11	4.8.8.2
Watertightness	3.12	4.8.8.3

4.5 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification

b. Drawing and specification number and date, together with identification and date of changes thereto

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review

d. Number of items in the lot

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this

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authority is vested in another agent of the certifying organization.

4.6 Inspection provisions for production.

4.6.1 Lot formation. In this specification the term inspection lot refers to an essentially homogeneous collection of units or product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specifications, or revision thereof. Changes to either the process, specification or drawing not affecting safety, performance, interchangability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall not be less than the smallest estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.6.2 Examination. The examinations specified herein shall be performed on inspection lots (see 4.6.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent.

a. *Sampling Plans.* Sampling plans shall be in accordance with tables II through VIII of this specification. Sampling plans for major defects shall be in accordance with sampling plans specified herein, except that variable sampling plans in accordance with MIL-STD-414 may be used if requested by the contractor and approved by the contracting officer.

b. *Critical Defects.* Unless otherwise specified, inspection for critical defects shall be 100 percent.

c. *Major Defects.* When required, major defects shall be inspected 100 percent. Except where 100 percent inspection is specified, inspection for major defects shall be to the individual sampling plan tables indicated in the classification of defects and test paragraphs provided in this detail specification. The sampling plans for testing (see 4.8) shall be as specified.

d. *Unlisted Defects.* See MIL-A-2550.

e. *Individual Defects.* Instead of applying the sampling plans designated herein to the individual defects, the group sampling plans listed for each major subparagraph in 4.6.2 may be used at the option of the contractor for those groups where a group sampling plan is assigned. Each group sampling plan applies to a major group indicated in the subparagraphs of paragraph 4.6.2, and if this method is elected, it shall be used for all of the subparagraphs of paragraph 4.6.2, where group sampling plans are so designated for this purpose.

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TABLE II. Sampling plan I.

Lot size	Sample size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE III. *Sampling plan II.*

Lot size	Sample size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE IV. *Sampling plan III.*

Lot size	Sample size
2 to 8	All
9 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38
10,001 to 35,000	46
35,001 to 150,000	56
150,001 and over	64
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE V. *Sampling plan IV.*

Lot size	Sample size
2 to 80	All
81 to 1200	80
1201 to 3200	120
3201 to 35,000	189
35,001 to 150,000	218
150,001 to 500,000	270
500,001 and over	303
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE VI. *Sampling plan V.*

Lot size	Sample size
2 to 200	All
201 to 10,000	200
10,001 to 35,000	300
35,001 to 500,000	476
500,001 and over	556
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE VII. *Sampling plan VI.*

Lot size	Sample size
2 to 125	All
126 to 3200	125
3201 to 10,000	192
10,001 to 35,000	294
35,001 to 150,000	294
150,001 to 500,000	345
500,001 and over	435
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE VIII. *Sampling plan VII.*

Lot size	Sample size
2 to 800	All
801 to 500,000	800
500,001 and over	1200
In all cases: Acceptance number is ZERO Rejection number is ONE	

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4.7 Classification of defects. Each sample shall be tested for defects as specified in tables IX through XIV.

TABLE IX. *Classification of defects, squib Mark 1 Mod 0 (Drawing 233AS522).*

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:			
1	Bottom of case deformed, split or cracked	Visual	100%
2	Lead wire ends not short circuited	Visual	100%
Major:			
101	Case crimp not continuous or not uniform	Visual	III
102	Shellac finish incomplete	Visual	III
103	Outside diameter nearest crimp	Gauge	III
104	Length of bare lead wire from insulation to shunting connector	Gauge	III
105	Length of exposed insulation lead	Gauge	III
106	Length forward end of case to base of booster charge assembly	Gauge	III
107	Electrical circuit and continuity test	Igniter Circuit Tester	IV
108	Dielectric test	Microampere Megohmeter	IV
109	No-fire sensitivity	Data Acq. Firing Bay	V
110	Firing time test	Data Acq. Firing Bay	I
111	Output limits	Bomb test fixture	I

TABLE X. *Classification of defects, Plug Assembly (Drawing 233AS532).*

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:	None defined		
Major:			
101	Lead wire insulation charred or otherwise damaged after molding	Visual	I
102	Sprue cavity extends to bare lead wire or permits insulation to move away from plug	Visual	I
103	Pull Test	Gauge	VI
104	Dielectric Test on wires	Microampere Megohmeter	VI

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TABLE XI. *Classification of defects, Initiation Assembly (Drawing 233AS533).*

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:			
1	Initiation bead contacts plug surface between terminals	Visual	100%
Major:			
101	Cellulose lacquer coating incomplete	Visual	II
102	Initiation charge projects beyond the .192 diameter	Gauge	II

TABLE XII. *Classification of defects, Casing Assembly (Drawing 233AS530).*

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:	None defined		
Major:			
101	Ledge on end of case	Visual	I
102	Smallest diameter at open end	Gauge	I
103	Corner section less than .004 inches thick	Gauge	I
104	Thickness of side section	Gauge	I

TABLE XIII. *Classification of defects, Spacer Assembly* (Drawing 233AS531).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:	None defined		
Major:			
101	Inside diameter of spacer	Gauge	II

TABLE XIV. *Classification of defects, Bridge Wire Assembly (Drawing 233AS534).*

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical:	None defined		
Major:			
101	Load test	Scale	VII

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4.8 Testing.

4.8.1 Electrical circuit and continuity test, major defect No. 107 table IX (see Drawing 233AS522). Each lot of squibs shall be tested as specified in 4.9. The sampling plan for this test shall be in accordance with table V sampling plan IV. One defective item not meeting the requirements of 3.3 shall be cause for lot rejection.

4.8.2 Dielectric test, major defect No. 109 (see table IX). Using the acceptable samples from 4.8.1, test the squibs as specified in 4.9.2. The sampling plan for this test shall be in accordance with table V, sampling plan IV. One defective item not meeting the requirements of 3.4 shall be cause for lot rejection. This is considered a destructive test. Any item so tested shall not be returned to the lot.

4.8.3 No-fire sensitivity test, major defect No. 110 (see table IX). Each lot of squibs shall be tested as specified in 4.9.3. The sampling plan for this test shall be in accordance with table VI, sampling plan V. One defective sample failing the requirements of 3.5 shall be cause for rejection. This is considered to be a destructive test. Any item so tested shall not be returned to the lot.

4.8.4 Firing time test, major defect No. 111 (see table IX). Each lot of squibs shall be tested as specified in 4.9.4. The sampling plan for this test shall be in accordance with table II, sampling plan I. If the sample fails the requirements of 3.6, or has one or more duds or misfires, the lot shall be rejected.

4.8.5 Pull test (see Drawing 233AS532), major defect No. 103 (see table X). Each lot of plug assemblies shall be tested as specified in 4.9.5. The sampling plan for this test shall be in accordance with table VII, sampling plan VI. One defective sample failing the requirements of 3.7 shall be cause for lot rejection.

4.8.6 Load test (see Drawing 233AS534), major defect No. 101 (see table XIV). Each lot of bridge wire assemblies shall be tested as specified in 4.9.6. The sampling plan for this test shall be in accordance with table VIII, sampling plan VII. One defective sample failing the requirements of 3.8 shall be cause for lot rejection.

4.8.7 Output limits test (see 3.9), major defect No. 111 (see table IX). Each lot of squibs shall be tested as specified in 4.9.7. The sampling plan for this test shall be in accordance with table II, sampling plan I. If one sample fails the requirements of 3.9, the lot shall be rejected.

4.8.8 Special tests. The following tests are to be performed on the first article sample and on each rejected lot resubmitted for acceptance in addition to the normal acceptance tests. Information as to sample size and test sequence is given in figure 1.

4.8.8.1 Jolt test (see 3.10). The test sample shall be tested as specified in 4.9.8.1. Thereafter, the squibs shall meet the requirements of 3.10.

4.8.8.2 Vibration test (see 3.11). The test sample shall be tested as specified in 4.9.8.2. Thereafter, the squibs shall meet the requirements of 3.11.

4.8.8.3 Watertightness test (see 3.12). The test sample shall be tested as specified in 4.9.8.3. Thereafter, the squibs shall meet the requirements of 3.12.

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4.9 Test methods and procedures.

4.9.1 Electrical circuit and continuity test (see 4.8.1). The electrical circuit of the squib between the lead wires shall be continuous and shall be within the resistance limits of 0.8 to 1.2 ohms inclusive. The equipment used in measuring the igniter electrical continuity and resistance shall not subject the squib firing circuit to more than 5 milliamperes.

4.9.2 Dielectric test (see 4.8.2). The resistance between the connected lead wires, and any portion of the bare metal exposed on the squib case, shall be not less than 50 megohms. The test potential shall be either 1000 ± 100 volts, 60 Hz alternating current (ac) or 500 ± 50 volts dc. The test potential shall be applied for not less than one second.

4.9.3 No-fire sensitivity test (see 4.8.3). The electrical circuit of the squib between the lead wires shall not fire when energized with 0.20 amperes, 24 volts dc for 5.25 ± 0.25 seconds.

4.9.4 Firing time test (see 4.8.4). The squib shall fire within the limits specified when energized with 1.5 ± 0.15 amperes at 70 ± 5 °F (21.1 °C ± 2.8 °C) using test equipment conforming to paragraph 4.9.7. The squib shall be fired using a dc constant current source. The firing current pulse shall have a rise time of 0.2 milliseconds or less from 10% to 90%. The firing current shall not exhibit any switch bounce or ripple outside of the 1.35 to 1.65 ampere limits and shall settle within 0.5 millisecond when applied to a 1.0 Ohm resistor (Real Load). The firing time shall be measured with a minimum accuracy of ±0.1 milliseconds from first indication of applied current to first indication of pressure. The test equipment conforming to paragraph 4.9.7 shall be used to measure and record the current pulse and squib pressure.

4.9.5 Pull test (see 4.8.5). The plug assembly shall meet the requirements of 3.7 when the individual lead wires are pulled with a force of 10 to 10.1 lbs.

4.9.6 Load test (see 4.8.6). The bridge wire assembly shall meet the requirements of 3.8 when a load of 25 to 27 grams is applied normal to the axis of the bridge wire at its center.

4.9.7 Output limits test (see 4.8.7). Fifty percent of the squibs for first article testing and lot acceptance testing shall be tested at -50 °F (-46 °C) ± 5 °F and the remaining 50 % of the squibs shall be tested at +150 °F (+66 °C) ± 5 °F. The test shall be performed using a closed bomb test system as described in sections 4.9.7.1 through 4.9.7.11. A procedure shall be submitted that ensures the squib temperature at the time of firing is as close to the conditioning temperatures as practical and must be approved by the contracting activity. The 10 cc closed bomb shall be cleaned between each shot to assure that the 10 cc volume is maintained for each test. The contractor shall provide, assemble, disassemble, calibrate, maintain, and have the system approved by the procuring activity, in accordance with the following requirements.

4.9.7.1 Squib closed bomb fixture. Squib closed bomb assembly NAVAIR Drawing 233AS814.

4.9.7.2 Pressure transducers. Standard commercial strain gauge type pressure transducers are adequate if they meet the following specifications:

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a.	Rated pressure:	0–1000 lb/in^2
b.	Electrical circuit:	strain gauge
c.	Physical type:	diaphragm
d.	Compensated temperature range:	-20 to 180 °F
e.	Calibration accuracy:	±0.25% rated output (RO)
f.	Nonlinearity + repeatability + hysteresis ±0.25% RO	
	1) Nonlinearity:	0.15% RO
	2) Repeatability:	0.05% RO
	3) Hysteresis:	0.05% RO
g.	Temperature effect on span and zero:	0.0015% per °F
h.	Natural frequency	17 kHz minimum

4.9.7.3 Calibration verification. Double shunt (opposing legs) calibration shall be done and recorded prior to every firing. The calibration before each test entails shunting resistors across the wheatstone bridge to simulate known pressure levels. Transducer calibration shall be done at least every six months according to the National Institute of Standards and Technology.

4.9.7.4 Signal conditioner. Commercial units are available that will adequately perform the task. Signal conditioners shall provide excitation voltage, bridge (zero) balance, and four-step double shunt calibration. An eight-wire transducer interface shall be used; four wires for calibration and four wires for excitation and signal. All units should meet the following specifications:

a.	Excitation voltage:	0–15 Volts direct current (Vdc)
b.	Regulation (line):	0.01%
	(load):	0.01%
c.	Ripple:	<100 micro-volts peak-peak
d.	Resolution:	5 milli-volts
e.	Isolation:	10,000 mega ohms
f.	Temperature coefficient:	0.005 %/°F
g.	Low pass frequency response:	50 kHz minimum

4.9.7.5 Amplifier. A standard commercial unit is suggested with the following requirements:

a.	Gain step accuracy:	± 0.1%
b.	Gain temperature coefficient:	± 0.005 % FS/°F
c.	Gain stability:	± 0.005 % FS at constant temperature
d.	Linearity:	± 0.005 % FS
e.	Bandwidth (3 dB point):	dc to 100 kHz

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f. Noise full bandwidth: 6 micro-volts reference-to-input +6 milli-volts reference-to-output (RMS)

g. Setting time: 30 micro-sec to within ± 0.1%

4.9.7.6 Filter. An anti-aliasing filter with 3 dB cutoff at 3000 Hz shall be used or shall be built into the amplifier or digitizer and implemented. The recommended type is a critically damped, bessel or butterworth two-pole filter. The filter shall exhibit no greater than 3% overshoot to an impulse. The filter shall not produce greater than a 0.1 millisecond phase shift to a 1000 Hz wave.

4.9.7.7 Digitizer. The setup will use at least a 12-bit digitizer and take 10,000 samples per second minimum for 100 milliseconds minimum.

4.9.7.8 Computer/storage. The computer shall be interfaced to the digitizer and, at a minimum, be able to receive the data from calibration and squib tests, store the data in non-volatile memory, locate the maximum pressure and its corresponding time, and determine the firing time. The computer shall be reliable and not experience errors repeatedly.

4.9.7.9 Printer. The printer shall be interfaced to the computer and shall be capable of producing data plots as well as numeric tabular data lists in a quick and clear fashion.

4.9.7.10 System Configuration. The system consisting primarily of transducers, signal conditioners, amplifiers, filters, digitizers, connectors, wiring, and enclosures shall be configured as no less than a two-channel system. One channel shall record pressure and one channel shall record firing current. Standard instrumentation practices shall be utilized to assure that noise, glitches, crosstalk or other factors do not interfere with the data integrity. The overall system shall exhibit an accuracy of less than or equal to 0.25% error. Channel-to-channel phase shift shall be less than 0.1 milliseconds. Multiple components from various commercial vendors are available as well as fully integrated systems in a single enclosure that will perform adequately.

4.9.7.11 Software. The software, as a minimum, shall provide the information to accurately determine the firing time, the maximum pressure, and the time from first application of current to the maximum pressure. The software shall adequately capture and process the required data. The software shall operate the computer in a glitch-free fashion. Lock-up, errors, loss of data, or other problems shall not occur on a repeated basis.

4.9.8 Special test procedures.

4.9.8.1 Jolt test (see 4.8.8.1). The squib shall be subjected to the jolt test in accordance with MIL-STD-331 Test Number 101.

4.9.8.2 Vibration test (see 4.8.8.2). The squib shall be subjected to the transportation -vibration test in accordance with MIL-STD-331 Test Number 123.

4.9.8.3 Watertightness test (see 4.8.8.3). The squibs shall be subjected to a cycle of four pressure

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conditions for a period of 3 to 5 minutes each. The cycle shall be repeated 21 times continuously. Procedures are as follows:

a. Submerge squibs in 71.1 ± 5.6 °C water at absolute pressure of approximately 58 inches of mercury (inHg).

b. Drain or pump water from chamber

c. Reduce chamber absolute pressure to not greater than 5 in Hg

d. Fill chamber with tap water to an absolute pressure of approximately 58 in Hg. Temperature differential between tap water and hot water, condition (a), shall be not greater than 32.2 °C.

e. Drain or pump water from chamber

f. Reduce chamber absolute pressure to not greater than three in Hg.

4.10 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-1169.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The squib covered by this specification is intended for use as in an igniter for the rocket motors in the 2.75-inch rocket.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2)

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c. When first article is required (see 3.1 and 4.3)

d. Drawings when required, providing dimensions, tolerances, hardness, and other design details (see 3.2)

e. First article sample size (see 4.3.1)

f. Conditions for quality conformance inspection (see 4.4)

g. Requirements for preservation, packing and marking (see section 5)

h. Requirements for defect reporting (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Defect reporting. From each lot produced, the contractor should submit to the address specified in the contract and in accordance with DD Form 1423, the results of the tests specified below within three days after completion of acceptance testing of the lot. The test results are to include data as indicated in Table XV. A failure analysis is required whenever a unit does not meet the requirements for acceptance of the tests listed in Table XV. The failure analysis should contain specific readings for those units which do not meet the specification requirements as well as additional pertinent information. Submission of these reports should be continued for each lot until the contractor has produced 250,000 units in lots accepted without waiver. After this 250,000 units has been produced, a report for every fifth lot for the duration of the contract should be required. In addition, a report should be submitted for every lot that fails to meet the specification requirements. The defect report cycle should be repeated whenever a first article submission is required or as directed by the Government procuring activity.

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Charge, explosive
Explosives
Igniter
Rocket assemblies
Support mechanism

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TABLE XV. *Failure analysis testing.*

Date _____		Lot No. _____ Accepted _____ with waiver _____ without waiver
Electrical circuit and continuity test (see 4.8.1)		
No. of units tested	No. of units failed	Failure analysis
Dielectric test (see 4.8.2)		
No. of units tested	No. of units failed	Failure analysis
No-fire sensitivity test (see 4.8.3)		
No. of units tested	No. of units failed	Failure analysis
Firing time test (see 4.8.4)		
No. of units tested	No. of units failed	Failure analysis
Pull test (see 4.8.5)		
No. of units tested	No. of units failed	Failure analysis
Load test (see 4.8.6)		
No. of units tested	No. of units failed	Failure analysis
Output limits test (see 4.8.7)		
No. of units tested	No. of units failed	Failure analysis

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. Authority for changes can be found in Engineering Change Proposals M2R3052 and M5R4054.

Preparing activity:
Navy – OS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-14977C(OS)	2. DOCUMENT DATE (YYMMDD) 970618

3. DOCUMENT TITLE
SQUIB, MARK 1 MOD 0 FOR 2.75 INCH ROCKET

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8410T)	b. TELEPHONE (Include Area Code) (1) Commercial: 301-743-4700 (2) AUTOVON: 354-4700
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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